

Exp No :1 Setting up Hyperledger Fabric Development Environment

Date:17/07/2026

Aim:

To install and configure the **Hyperledger Fabric development environment on Linux**, create the Fabric test network, and create an application channel named **mychannel** .

Requirements:

- Linux Operating System (Kali)
- Docker and Docker Compose
- Go Programming Language
- Git and Curl
- Hyperledger Fabric Samples, Binaries and Docker Images

Procedure:

Step 1: Update the System

Open the Linux/Kali terminal and update the system packages.

```
sudo apt update  
sudo apt upgrade -y
```

Step 2: Verify Docker Installation

Check whether Docker is installed.

```
docker --version
```

Start and enable Docker:

```
sudo systemctl start docker  
sudo systemctl enable docker
```

Test Docker:

```
docker run hello-world
```

If you see “**Hello from Docker!**”, Docker is working correctly.

```
(Naveen@LAPTOP-258I2DTL)-[~]
$ docker --version
Docker version 29.7.2, build a7dcaa6

(Naveen@LAPTOP-258I2DTL)-[~]
$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash
```

Step 3: Verify Go Installation

Check the installed Go version:

```
go version
```

```
(Naveen@LAPTOP-258I2DTL)-[~]
$ go version
go version go1.26.4 linux/amd64
```

Shows Go successfully installed and verified.

Step 4: Create Fabric Workspace

Create a directory for Hyperledger Fabric:

```
mkdir -p ~/fabric-dev
cd ~/fabric-dev
```

```
(Naveen@LAPTOP-258I2DTL)-[~]
$ mkdir -p ~/fabric-dev

(Naveen@LAPTOP-258I2DTL)-[~]
$ cd fabric-dev/

(Naveen@LAPTOP-258I2DTL)-[~/fabric-dev]
$ pwd
/home/Naveen/fabric-dev
```

Step 5: Download Hyperledger Fabric

Download the Fabric installation script:

```
curl -sSL
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-
fabric.sh -o install-fabric.sh
```

Give execute permission:

```
chmod +x install-fabric.sh
```

```
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ curl -sSL https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh -o install-fabric.sh
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ chmod +x install-fabric.sh
```

Install Fabric Docker images, samples, and binaries:

```
./install-fabric.sh docker samples binary
```

```
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15868, done.
remote: Counting objects: 100% (2946/2946), done.
remote: Compressing objects: 100% (557/557), done.
remote: Total 15868 (delta 2570), reused 2389 (delta 2389), pack-reused 12922 (from 2)
Receiving objects: 100% (15868/15868), 24.11 MiB | 7.24 MiB/s, done.
Resolving deltas: 100% (9112/9112), done.
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch

Pull Hyperledger Fabric binaries

==> Downloading version 2.5.16 platform specific fabric binaries
==> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyp
==> Will unpack to: /home/Naveen/fabric-dev/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0         0     0      0     0      0      0      0      0     00:01      0
100 121.8M 100 121.8M     0      0  7.38M      0     00:16     00:16      9.26M
==> Done.
==> Downloading version 1.5.17 platform specific fabric-ca-client binary
==> Downloading: https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/
==> Will unpack to: /home/Naveen/fabric-dev/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0         0     0      0     0      0      0      0      0     00:01      0
100 31.65M 100 31.65M     0      0  5.17M      0     00:06     00:06      7.08M
==> Done.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
==> Pulling fabric Images
==> ghcr.io/hyperledger/fabric-peer:2.5.16
2.5.16: Pulling from hyperledger/fabric-peer
40d16f30db40: Pull complete
f2df6aa6a776: Pull complete
b7fdc54bd9fd: Pull complete
```

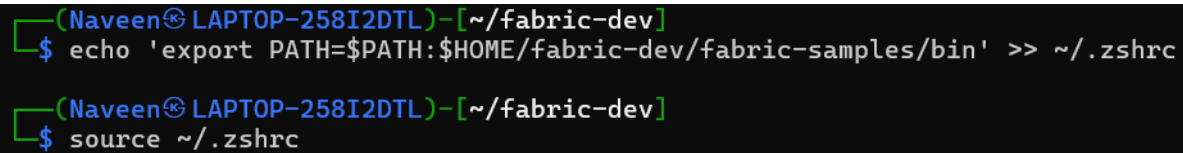
Step 6: Configure PATH

Add Fabric binaries to the PATH:

```
echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc
```

Reload the configuration:

```
source ~/.zshrc
```



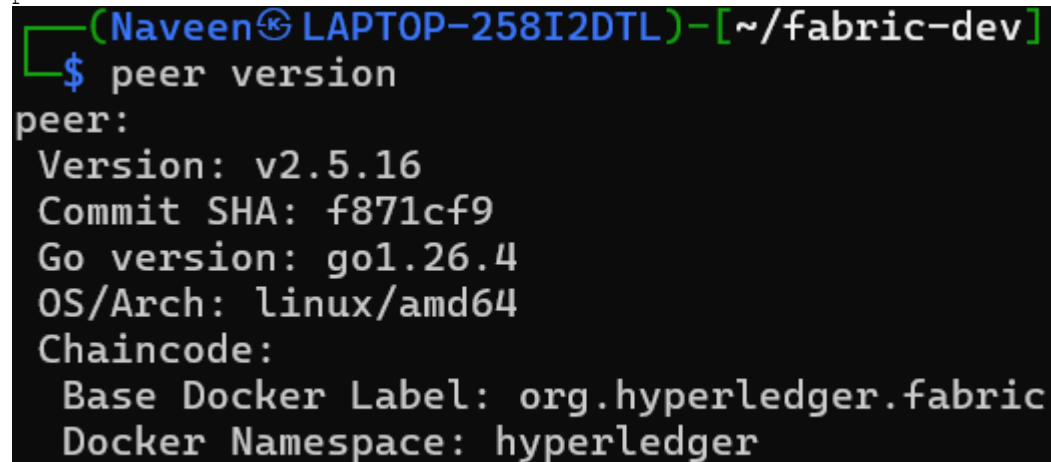
```
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc

(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ source ~/.zshrc
```

Step 7: Verify Fabric Binaries

Check the Fabric peer version:

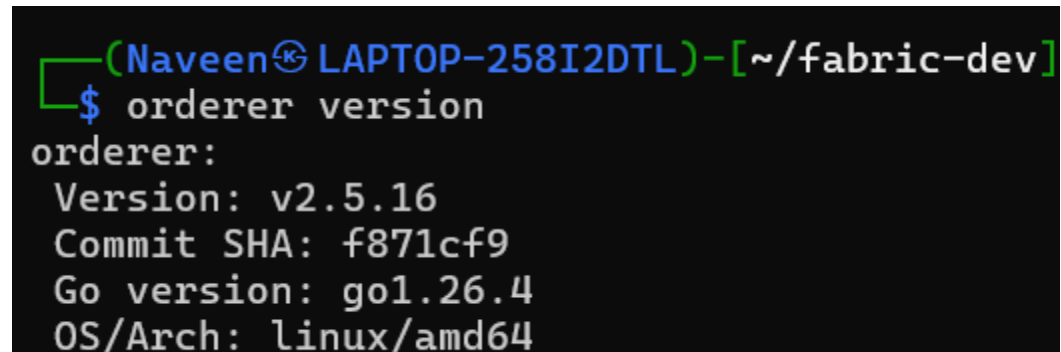
```
peer version
```



```
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ peer version
peer:
  Version: v2.5.16
  Commit SHA: f871cf9
  Go version: go1.26.4
  OS/Arch: linux/amd64
  Chaincode:
    Base Docker Label: org.hyperledger.fabric
    Docker Namespace: hyperledger
```

Check the orderer version:

```
orderer version
```

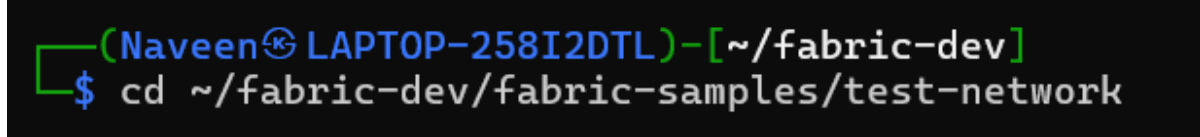


```
(Naveen@LAPTOP-258I2DTL)~[~/fabric-dev]
$ orderer version
orderer:
  Version: v2.5.16
  Commit SHA: f871cf9
  Go version: go1.26.4
  OS/Arch: linux/amd64
```

Step 8: Go to the Fabric Test Network

Navigate to the test-network directory:

```
cd ~/fabric-dev/fabric-samples/test-network
```

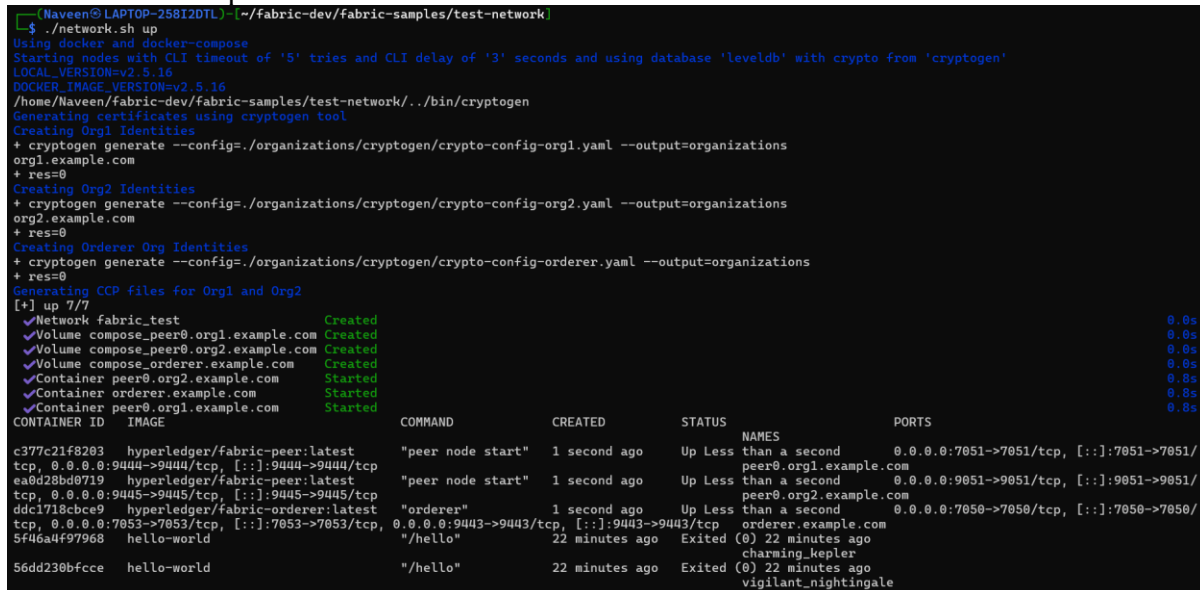


Step 9: Start the Fabric Network

Start the Fabric test network:

```
./network.sh up
```

This starts the required Fabric containers.

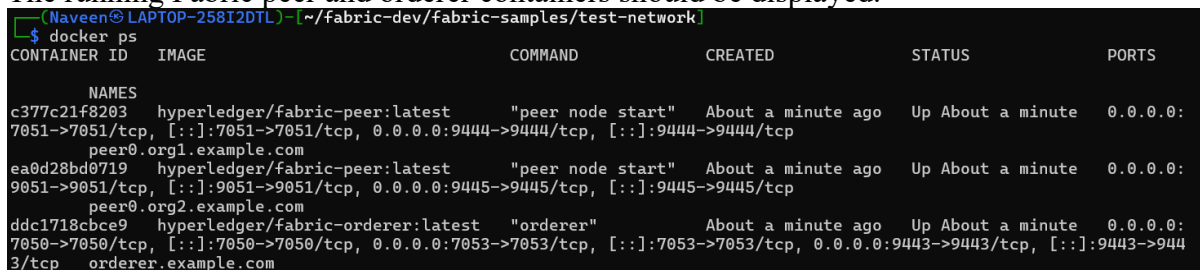


Step 10: Verify Running Containers

Check the running Docker containers:

```
docker ps
```

The running Fabric peer and orderer containers should be displayed.



Step 11: Create the Application Channel

Create a channel named **mychannel**:

```
./network.sh createChannel -c mychannel
```

The channel is created and the peer nodes join the channel.

```
(Naveen@LAPTOP-258I2DTL)~/fabric-dev/fabric-samples/test-network
$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] up 3/3
  ✓Container peer0.org1.example.com Running
  ✓Container peer0.org2.example.com Running
  ✓Container orderer.example.com Running
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
c377c21f8203	hyperledger/fabric-peer:latest	"peer node start"	2 minutes ago	Up 2 minutes
ea0d28bd0719	hyperledger/fabric-peer:latest	"peer node start"	2 minutes ago	Up 2 minutes
ddc1718cbce9	hyperledger/fabric-orderer:latest	"orderer"	2 minutes ago	Up 2 minutes
9443->9443/tcp	orderer.example.com			
5f46a4f97968	hello-world	"/hello"	24 minutes ago	Exited (0) 24 minutes ago
56dd230bfcce	hello-world	"/hello"	24 minutes ago	Exited (0) 24 minutes ago

```
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/Naveen/fabric-dev/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-08-17 09:22:06.652 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-08-17 09:22:06.657 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type
2026-08-17 09:22:06.658 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.Etcd
terval_size:16777216
2026-08-17 09:22:06.658 IST 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/Na
2026-08-17 09:22:06.662 IST 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-08-17 09:22:06.662 IST 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel gene
2026-08-17 09:22:06.662 IST 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
  "name": "mychannel",
  "url": "/participation/v1/channels/mychannel",
  "consensusRelation": "consenter",
  "status": "active",
  "height": 1
}
Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-08-17 09:22:12.814 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-08-17 09:22:12.847 IST 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Joining org2 peer to the channel...
Using organization 2
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-08-17 09:22:15.914 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-08-17 09:22:15.941 IST 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
```

Step 12: Stop the Fabric Network

After completing the experiment, stop the network:

```
./network.sh down
```

```
(Naveen@LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/test-network]
$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 7/7
✓Container peer0.org2.example.com      Removed
✓Container peer0.org1.example.com      Removed
✓Container orderer.example.com         Removed
✓Volume compose_orderer.example.com    Removed
✓Network fabric_test                   Removed
✓Volume compose_peer0.org2.example.com Removed
✓Volume compose_peer0.org1.example.com Removed
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
b05093807bb0: Pull complete
7270b3e1860c: Download complete
Digest: sha256:dc2d74b28e4cf8984fa52af1f39bc7c3d9c73760b41a74d629f5d11b1ab28616
Status: Downloaded newer image for busybox:latest
```

Result

The Hyperledger Fabric development environment was successfully configured. The Fabric test network was started, **mychannel** was successfully created, and the peer nodes joined the channel successfully.